

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
LLM & Agents	LLM agents, tool use, multi-stage reasoning, MCP (basics), prompt design, provider routing, OpenAI / Groq APIs, Ollama, HuggingFace Transformers
ML & Research	PyTorch, JAX, NumPy, mechanistic interpretability / activation steering, variational inference, autodiff, generative models, statistical evaluation (bootstrap CIs, McNemar)
Retrieval & Eval	RAG, multi-hop QA, RRF, cross-encoder reranking, MMR, evaluation methodology, anti-Goodharting, holdout protection, sentence-transformers
Cloud & Data	AWS (S3, KMS), GCP, Azure, Docker, Git, CI/CD, Apache Spark / PySpark, PostgreSQL, MongoDB, Redis

Research & Projects

AgenticSearch - LLM Agent System

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Designed an **LLM agent architecture** for open-ended discovery: typed retrieval planning, tool use (search + rerank), evidence-regime classification, deterministic extraction with LLM fallback, and output verification, returning **cell-level provenance** for every value.
- Hardened the agent with **150+ automated tests** and provider routing/fallback across LLM backends, bridging research-grade reasoning with production-grade reliability.

AutoEval - Self-Improving Evaluation Agent

github.com/Deva-1903/AutoEval

Python, LLM Evaluation, Git Automation

- Designed an agent that iteratively **self-improves** an evaluation suite, proposing test cases and labels and accepting only updates that raise **discriminative power** on a holdout-heavy objective, with anti-Goodharting guardrails for an auditable, reproducible workflow.

Refusal Decay - LLM Alignment & Interpretability

UMass CS 602 Research Project (graded full marks)

PyTorch, HuggingFace Transformers, Llama-3.1-8B-Instruct

- Ran a mechanistic-interpretability study of how prefilling attacks weaken refusal in **Llama-3.1-8B**; extracted the refusal direction and **steered model behavior** via activation patching and direction injection, reporting a **clean null** (0/300) with multi-seed controls, bootstrap 95% CIs, and McNemar’s test.

Generative Models & Optimization

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented a reverse-mode **autodiff engine from scratch** (validated against JAX) and built a normalizing flow and **DDPM** from scratch in JAX; derived GD/SGD convergence guarantees and benchmarked 5 architectures across 3 optimizers.

KG2RAG-Enhanced - Multi-Hop QA & Retrieval

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, PyTorch | UMass CS 685 Advanced NLP (team)

- Led the **multi-view retrieval** component of a team KG-guided RAG pipeline (sub-question decomposition, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; replaced spreadsheet workflows and accelerated iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.